

XCS221 Assignment 6 — Car Tracking

Due Sunday, February 6 at 11:59pm PT.

Guidelines

1. These questions require thought, but do not require long answers. Please be as concise as possible.
2. If you have a question about this homework, we encourage you to post your question on our Slack channel, at <http://xcs229ii-scpd.slack.com/>
3. Familiarize yourself with the collaboration and honor code policy before starting work.
4. For the coding problems, you may not use any libraries except those defined in the provided started code. In particular, ML-specific libraries such as `scikit-learn` are not permitted.

Submission Instructions

Written Submission: All students must submit an electronic PDF containing solutions to the written questions. As long as the submission is legible and well-organized, the course staff has no preference between a handwritten and a typeset $\text{\LaTeX}$ submission. Students wishing to typeset their documents should follow these recommendations:

- Type responses only in the `*-sol.tex` files.
- Use the commented recommendations within the Makefile to get started.

Coding Submission: All assignment code is in the `src/` subdirectory. You will submit only the `src/submission.py` file. Please only make changes between the lines containing `### START_CODE_HERE ###` and `### END_CODE_HERE ###`. Do not make changes to files other than `src/submission.py`.

The unit tests in `src/grader.py` will be used to autograde your submission. Run the autograder locally using the following terminal command within the `src/` subdirectory:

```
python grader.py
```

There are two types of unit tests used by our autograders:

- **basic:** These unit tests will verify only that your code runs without errors on obvious test cases.
- **hidden:** These unit tests will verify that your code produces correct results on complex inputs and tricky corner cases. In the student version of `src/grader.py`, only the setup and inputs to these unit tests are provided. When you run the autograder locally, these test cases will run, but the results will not be verified by the autograder.

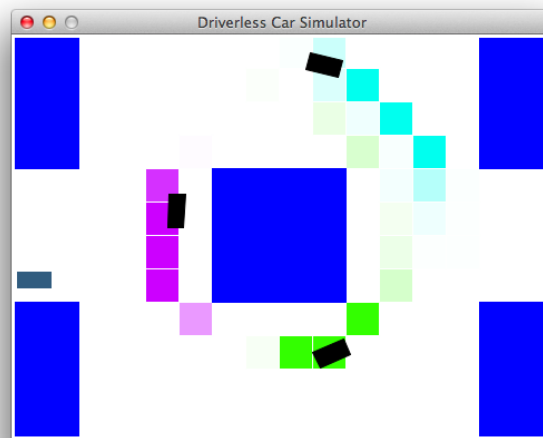
For debugging purposes, a single unit test can be run locally. For example, you can run the test case `3a-0-basic` using the following terminal command within the `src/` subdirectory:

```
python grader.py 3a-0-basic
```

Before beginning this course, we highly recommend you walk through our Anaconda tutorial **NO LINK YET** to familiarize yourself with our coding environment. Please use the env defined in `src/environment.yml` to run your code. This is the same environment used by our autograder.

Honor code

We strongly encourage students to form study groups. Students may discuss and work on homework problems in groups. However, each student must write down the solutions independently, and without referring to written notes from the joint session. In other words, each student must understand the solution well enough in order to reconstruct it by him/herself. In addition, each student should write on the problem set the set of people with whom s/he collaborated. Further, because we occasionally reuse problem set questions from previous years, we expect students not to copy, refer to, or look at the solutions in preparing their answers. It is an honor code violation to intentionally refer to a previous year's solutions.



Introduction

This assignment is a modified version of the Driverless Car assignment written by Dr. Chris Piech.

A study by the World Health Organization found that road accidents kill a shocking 1.24 million people a year worldwide. In response, there has been great interest in developing autonomous driving technology that can drive with calculated precision and reduce this death toll. Building an autonomous driving system is an incredibly complex endeavor. In this assignment, you will focus on the sensing system, which allows us to track other cars based on noisy sensor readings.

Getting started.

You will be running two files in this assignment - `grader.py` and `drive.py`. The `drive.py` file is not used for any grading purposes, it's just there to visualize the code you will be writing and help you gain an appreciation for how different approaches result in different behaviors (and to have fun!). Let's start by trying to drive manually.

```
python drive.py -l lombard -i none
```

You can steer by either using the arrow keys or 'w', 'a', and 'd'. The up key and 'w' accelerates your car forward, the left key and 'a' turns the steering wheel to the left, and the right key and 'd' turns the steering wheel to the right. Note that you cannot reverse the car or turn in place. Quit by pressing 'q'. Your goal is to drive from the start to finish (the green box) without getting in an accident. How well can you do on the windy Lombard street without knowing the location of other cars? Don't worry if you're not very good; the teaching staff were only able to get to the finish line 4/10 times. An accident rate of 60% is pretty abysmal, which is why we're going to use AI to do this.

Flags for `python drive.py`:

- `-a`: Enable autonomous driving (as opposed to manual).
- `-i <inference method>`: Use `none`, `exactInference`, `particleFilter` to (approximately) compute the belief distributions over the locations of the other cars.
- `-l <map>`: Use this map (e.g. `small` or `lombard`). Defaults to `small`.
- `-d`: Debug by showing all the cars on the map.
- `-p`: All other cars remain parked (so that they don't move).

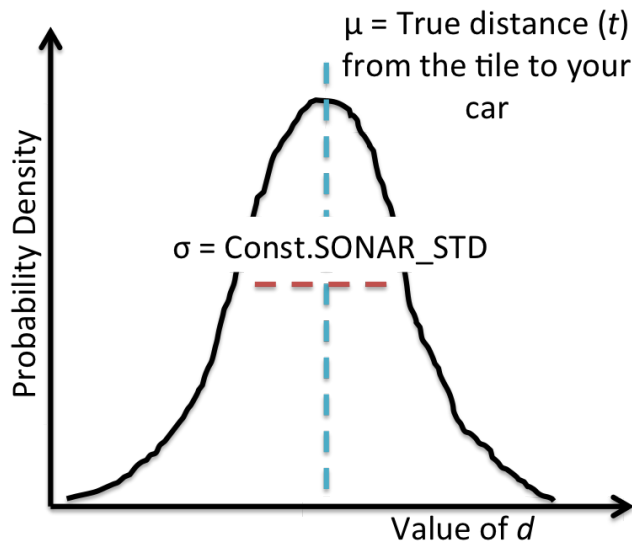
Modeling car locations.

We assume that the world is a two-dimensional rectangular grid on which your car and K other cars reside. At each time step t , your car gets a noisy estimate of the distance to each of the cars. As a simplifying assumption, we assume that each of the K other cars moves independently and that the noise in sensor readings for each car is also independent. Therefore, in the following, we will reason about each car independently (notationally, we will assume there is just one other car).

At each time step t , let $C_t \in \mathbb{R}^2$ be a pair of coordinates representing the *actual* location of the single other car (which is unobserved). We assume there is a local conditional distribution $p(c_t | c_{t-1})$ which governs the car's movement. Let $a_t \in \mathbb{R}^2$ be your car's position, which you observe and also control. To minimize costs, we use a simple sensing system based on a microphone. The microphone provides us with D_t , which is a Gaussian random variable with mean equal to the true distance between your car and the other car and variance σ^2 (in the code, σ is `Const.SONAR_STD`, which is about two-thirds the length of a car). In symbols,

$$D_t \sim \mathcal{N}(\|a_t - C_t\|, \sigma^2).$$

For example, if your car is at $a_t = (1, 3)$ and the other car is at $C_t = (4, 7)$, then the actual distance is 5 and D_t might be 4.6 or 5.2, etc. Use `util.pdf(mean, std, value)` to compute the probability density function (PDF) of a Gaussian with given mean `mean` and standard deviation `std`, evaluated at `value`. Note that evaluating a PDF at a certain value does not return a probability – densities can exceed 1 – but for the purposes of this assignment, you can get away with treating it like a probability. The Gaussian probability density function for the noisy distance observation D_t , which is centered around your distance to the car $\mu = \|a_t - C_t\|$, is shown in the following figure:



Your job is to implement a car tracker that (approximately) computes the posterior distribution $\mathbb{P}(C_t | D_1 = d_1, \dots, D_t = d_t)$ (your beliefs of where the other car is) and update it for each $t = 1, 2, \dots$. We will take care of using this information to actually drive the car (i.e., set a_t to avoid a collision with c_t), so you don't have to worry about that part.

To simplify things, we will discretize the world into **tiles** represented by `(row, col)` pairs, where $0 \leq \text{row} < \text{numRows}$ and $0 \leq \text{col} < \text{numCols}$. For each tile, we store a probability representing our belief that there's a car on that tile. The values can be accessed by: `self.belief.getProb(row, col)`. To convert from a tile to a location, use `util.rowToX(row)` and `util.colToX(col)`.

Here's an overview of the assignment components:

- Problem 1 (written) will give you some practice with probabilistic inference on a simple Bayesian network.
- In Problems 2 and 3 (code), you will implement `ExactInference`, which computes a full probability distribution of another car's location over tiles `(row, col)`.

- In Problem 4 (code), you will implement `ParticleFilter`, which works with particle-based representation of this same distribution.
- Problem 5 (written) gives you a chance to extend your probability analyses to a slightly more realistic scenario where there are multiple other cars and we can't automatically distinguish between them.

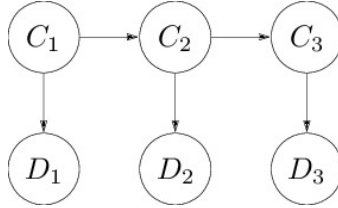
A few important notes before we get started:

- Past experience suggests that this will be one of the most conceptually challenging assignments of the quarter for many students. Please start early, especially if you're low on late days!
- We strongly recommend that you attend/watch the lectures on Bayesian networks and HMMs before getting started, and keep the slides handy for reference while you're working.
- The code portions of this assignment are short and straightforward – no more than about 30 lines in total – but only if your understanding of the probability concepts is clear! (If not, see the previous point.)
- As a notational reminder: we use the lowercase expressions $p(x)$ or $p(x|y)$ for local conditional probability distributions, which are defined by the Bayesian network. We use the uppercase expressions $\mathbb{P}(X = x)$ or $\mathbb{P}(X = x|Y = y)$ for joint and posterior probability distributions, which are not pre-defined in the Bayesian network but can be computed by probabilistic inference.

Please review the lecture slides for more details.

1. Bayesian Network Basics

First, let us look at a simplified version of the car tracking problem. For this problem only, let $C_t \in \{0, 1\}$ be the actual location of the car we wish to observe at time step $t \in \{1, 2, 3\}$. Let $D_t \in \{0, 1\}$ be a sensor reading for the location of that car measured at time t . Here's what the Bayesian network (it's an HMM, in fact) looks like:



The distribution over the initial car distribution is uniform; that is, for each value $c_1 \in \{0, 1\}$:

$$p(c_1) = 0.5.$$

The following local conditional distribution governs the movement of the car (with probability ϵ , the car moves). For each $t \in \{2, 3\}$:

$$p(c_t | c_{t-1}) = \begin{cases} \epsilon & \text{if } c_t \neq c_{t-1} \\ 1 - \epsilon & \text{if } c_t = c_{t-1}. \end{cases}$$

The following local conditional distribution governs the noise in the sensor reading (with probability η , the sensor reports the *wrong* position). For each $t \in \{1, 2, 3\}$:

$$p(d_t | c_t) = \begin{cases} \eta & \text{if } d_t \neq c_t \\ 1 - \eta & \text{if } d_t = c_t. \end{cases}$$

Below, you will be asked to find the posterior distribution for the car's position at the second time step (C_2) given different sensor readings.

Important:

For the following computations, try to follow the general strategy described in lecture (marginalize non-ancestral variables, condition, and perform variable elimination). Try to delay normalization until the very end. You'll get more insight than trying to chug through lots of equations.

- (a) **[1 point (Written)]** Suppose we have a sensor reading for the second timestep, $D_2 = 0$. Compute the posterior distribution $\mathbb{P}(C_2 = 1 | D_2 = 0)$. We encourage you to draw out the (factor) graph.
- (b) **[1 point (Written)]** Suppose a time step has elapsed and we got another sensor reading, $D_3 = 1$, but we are still interested in C_2 . Compute the posterior distribution $\mathbb{P}(C_2 = 1 | D_2 = 0, D_3 = 1)$. The resulting expression might be moderately complex. We encourage you to draw out the (factor) graph.
- (c) **[2 points (Written)]** Suppose $\epsilon = 0.1$ and $\eta = 0.2$.
 - i. Compute and compare the probabilities $\mathbb{P}(C_2 = 1 | D_2 = 0)$ and $\mathbb{P}(C_2 = 1 | D_2 = 0, D_3 = 1)$. Give numbers, round your answer to 4 significant digits.
 - ii. How did adding the second sensor reading $D_3 = 1$ change the result? Explain your intuition for why this change makes sense in terms of the car positions and associated sensor observations.
 - iii. What would you have to set ϵ to be while keeping $\eta = 0.2$ so that $\mathbb{P}(C_2 = 1 | D_2 = 0) = \mathbb{P}(C_2 = 1 | D_2 = 0, D_3 = 1)$? Explain your intuition in terms of the car positions with respect to the observations.

2. Emission Probabilities

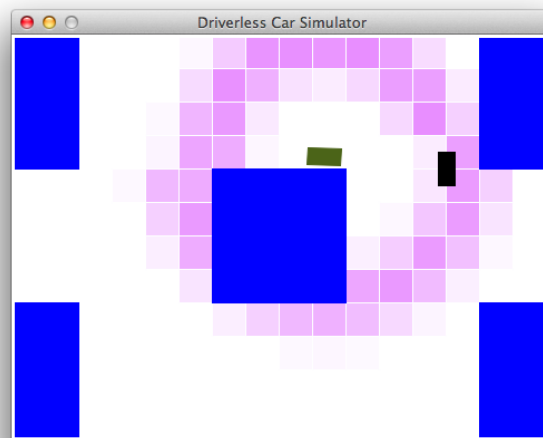
In this problem, we assume that the other car is stationary (e.g., $C_t = C_{t-1}$ for all time steps t). You will implement a function `observe` that upon observing a new distance measurement $D_t = d_t$ updates the current posterior probability from

$$\mathbb{P}(C_t \mid D_1 = d_1, \dots, D_{t-1} = d_{t-1})$$

to

$$\mathbb{P}(C_t \mid D_1 = d_1, \dots, D_t = d_t) \propto \mathbb{P}(C_t \mid D_1 = d_1, \dots, D_{t-1} = d_{t-1})p(d_t \mid c_t),$$

where we have multiplied in the emission probabilities $p(d_t \mid c_t)$ described earlier under “Modeling car locations”. The current posterior probability is stored as `self.belief` in `ExactInference`.



- (a) [5 points (Coding)] Fill in the `observe` method in the `ExactInference` class of `submission.py`. This method should modify `self.belief` in place to update the posterior probability of each tile given the observed noisy distance to the other car. After you're done, you should be able to find the stationary car by driving around it (using the flag `-p` means cars don't move):

Notes:

- You can start driving with exact inference now.

```
python drive.py -a -p -d -k 1 -i exactInference
```

You can also turn off `-a` to drive manually.

- It's generally best to run `drive.py` on your local machine, but if you do decide to run it on remote machine instead, please `ssh` with either the `-X` or `-Y` option in order to get the graphical interface; otherwise, you will get some display error message. Note: in this case, expect this graphical interface to be a bit slow... `drive.py` is not used for grading, but is just there for you to visualize and enjoy the game!
- Read through the code in `util.py` for the `Belief` class before you get started. You'll need to use this class for several of the code tasks in this assignment.
- Remember to normalize the posterior probability after you update it. (There's a useful function for this in `util.py`).
- On the small map, the autonomous driver will sometimes drive in circles around the middle block before heading for the target area. In general, don't worry too much about the precise path the car takes. Instead, focus on whether your car tracker correctly infers the location of other cars.
- Don't worry if your car crashes once in a while! Accidents do happen, whether you are human or AI. However, even if there was an accident, your driver should have been aware that there was a high probability that another car was in the area.

3. Transition Probabilities

Now, let's consider the case where the other car is moving according to transition probabilities $p(c_{t+1} | c_t)$. We have provided the transition probabilities for you in `self.transProb`. Specifically, `self.transProb[(oldTile, newTile)]` is the probability of the other car being in `newTile` at time step $t + 1$ given that it was in `oldTile` at time step t .

In this part, you will implement a function `elapseTime` that updates the posterior probability about the location of the car at a **current** time t

$$\mathbb{P}(C_t = c_t \mid D_1 = d_1, \dots, D_t = d_t)$$

to the **next** time step $t + 1$ conditioned on the same evidence, via the recurrence:

$$\mathbb{P}(C_{t+1} = c_{t+1} \mid D_1 = d_1, \dots, D_t = d_t) \propto \sum_{c_t} \mathbb{P}(C_t = c_t \mid D_1 = d_1, \dots, D_t = d_t) p(c_{t+1} \mid c_t).$$

Again, the posterior probability is stored as `self.belief` in `ExactInference`.

- (a) [6 points (Coding)] Finish `ExactInference` by implementing the `elapseTime` method. When you are all done, you should be able to track a moving car well enough to drive autonomously by running the following:

```
python drive.py -a -d -k 1 -i exactInference
```

Notes:

- You can also drive autonomously in the presence of more than one car:

```
python drive.py -a -d -k 3 -i exactInference
```

- You can also drive down Lombard:

```
python drive.py -a -d -k 3 -i exactInference -l lombard
```

On Lombard, the autonomous driver may attempt to drive up and down the street before heading towards the target area. Again, focus on the car tracking component, instead of the actual driving.

4. Particle Filtering

Though exact inference works well for the small maps, it wastes a lot of effort computing probabilities for *every available tile*, even for tiles that are unlikely to have a car on them. We can solve this problem using a particle filter. Updates to the particle filter have complexity that's linear in the number of particles, rather than linear in the number of tiles.

For a great conceptual explanation of how particle filtering works, check out this video on using particle filtering to estimate an airplane's altitude.

In this problem, you'll implement two short but important methods for the `ParticleFilter` class in `submission.py`. When you're finished, your code should be able to track cars nearly as effectively as it does with exact inference.

- (a) **[15 points (Coding)]** Some of the code has been provided for you. For example, the particles have already been initialized randomly. You need to fill in the `observe` and `elapseTime` functions. These should modify `self.particles`, which is a map from tiles (`row`, `col`) to the number of particles existing at that tile, and `self.belief`, which needs to be updated each time you re-sample the particle locations.

You should use the same transition probabilities as in exact inference. The belief distribution generated by a particle filter is expected to look noisier compared to the one obtained by exact inference.

```
python drive.py -a -i particleFilter -l lombard
```

To debug, you might want to start with the parked car flag (`-p`) and the display car flag (`-d`).

5. Which Car is it?

So far, we have assumed that we have a distinct noisy distance reading for each car, but in reality, our microphone would just pick up an undistinguished set of these signals, and we wouldn't know which distance reading corresponds to which car. First, let's extend the notation from before: let $C_{ti} \in \mathbb{R}^2$ be the location of the i -th car at the time step t , for $i = 1, \dots, K$ and $t = 1, \dots, T$. Recall that all the cars move independently according to the transition dynamics as before.

Let $D_{ti} \in \mathbb{R}$ be the noisy distance measurement of the i -th car at time step t , which is now not directly observed. Instead, we observe the **set** of distances $D_t = \{D_{t1}, \dots, D_{tK}\}$. (For simplicity, we'll assume that all distances are distinct values.) Alternatively, you can think of $E_t = (E_{t1}, \dots, E_{tK})$ as a list which is a uniformly random permutation of the noisy distances $(D_{t1}, \dots, D_{tK})$. For example, suppose $K = 2$ and $T = 2$. Before, we might have gotten distance readings of 1 and 2 for the first car and 3 and 4 for the second car at time steps 1 and 2, respectively. Now, our sensor readings would be permutations of $\{1, 3\}$ (at time step 1) and $\{2, 4\}$ (at time step 2). Thus, even if we knew the second car was distance 3 away at time $t = 1$, we wouldn't know if it moved further away (to distance 4) or closer (to distance 2) at time $t = 2$.

- (a) **[4 points (Written)]** Suppose we have $K = 2$ cars and one time step $T = 1$. Write an expression for the conditional distribution $\mathbb{P}(C_{11}, C_{12} \mid E_1 = e_1)$ as a function of the PDF of a Gaussian $p_{\mathcal{N}}(v; \mu, \sigma^2)$ and the prior probability $p(c_{11})$ and $p(c_{12})$ over car locations. Your final answer should not contain variables d_{11}, d_{12} .

Remember that $p_{\mathcal{N}}(v; \mu, \sigma^2)$ is the probability of a random variable, v , in a Gaussian distribution with mean μ and standard deviation σ .

Hint: for $K = 1$, the answer would be

$$\mathbb{P}(C_{11} = c_{11} \mid E_1 = e_1) \propto p(c_{11})p_{\mathcal{N}}(e_{11}; \|a_1 - c_{11}\|, \sigma^2).$$

where a_t is the position of the car at time t . You might find it useful to draw the Bayesian network and think about the distribution of E_t given $D_{t1}, \dots, D_{tK}$.

- (b) **[3 points (Written)]** Assuming the prior $p(c_{1i})$ of where the cars start out is the same for all i (i.e. for all K cars), show that the number of assignments for all K cars $(c_{11}, \dots, c_{1K})$ that obtain the maximum value of $\mathbb{P}(C_{11} = c_{11}, \dots, C_{1K} = c_{1K} \mid E_1 = e_1)$ is at least $K!$.

You can also assume that the car locations that maximize the probability above are unique ($C_{1i} \neq c_{1j}$ for all $i \neq j$).

Note: you don't need to produce a complicated proof for this question. It is acceptable to provide a clear explanation based on your intuitive understanding of the scenario.

- (c) **[1 point (Written)]** For general K , what is the treewidth corresponding to the posterior distribution over all K car locations at all T time steps conditioned on all the sensor readings:

$$\mathbb{P}(C_{11} = c_{11}, \dots, C_{1K} = c_{1K}, \dots, C_{T1} = c_{T1}, \dots, C_{TK} = c_{TK} \mid E_1 = e_1, \dots, E_T = e_T)?$$

Briefly justify your answer.

- (d) **[6 points (Written, Extra Credit)]** Now suppose you change your sensors so that at each time step t , they return the list of exact positions of the K cars, but shifted (with wrap around) by a random amount. For example, if the true car positions at time step 1 are $c_{11} = 1, c_{12} = 3, c_{13} = 8, c_{14} = 5$, then e_1 would be $[1, 3, 8, 5]$, $[3, 8, 5, 1]$, $[8, 5, 1, 3]$, or $[5, 1, 3, 8]$, each with probability $1/4$. Describe an efficient algorithm for computing $p(c_{ti} \mid e_1, \dots, e_T)$ for any time step t and car i . Your algorithm should not be exponential in K or T .